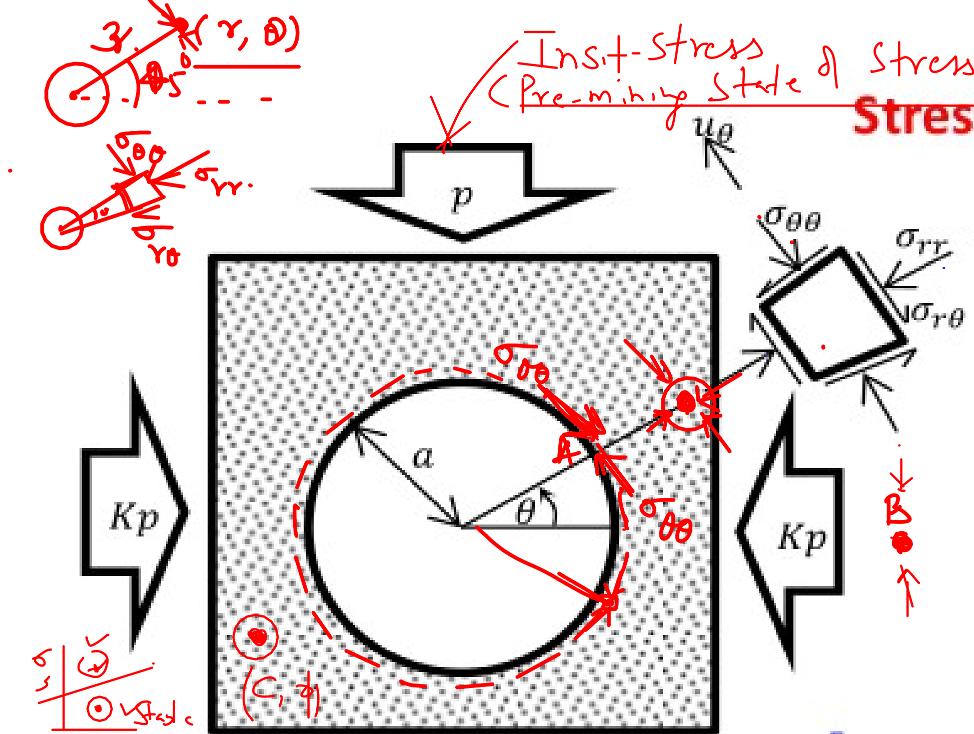
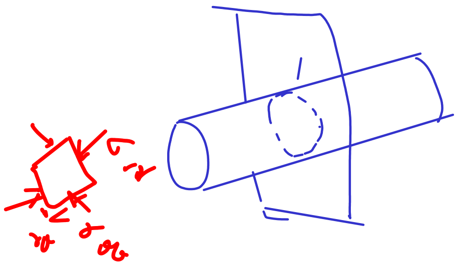


Stress and displacement distribution around a circular opening



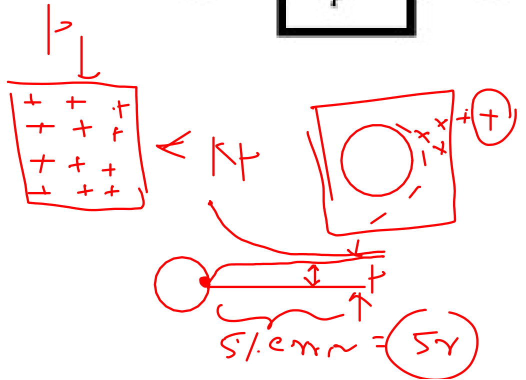
A	B
Circumference	Infinite dist
$\sigma_{\theta\theta}$ = ✓	$\sigma_{\theta\theta} = p$ ✓
$\sigma_{rr} = 0$	$\sigma_{rr} = kp$
$\sigma_{r\theta} = 0$	$\sigma_{r\theta} = 0$

Kirsch (1898) Equations

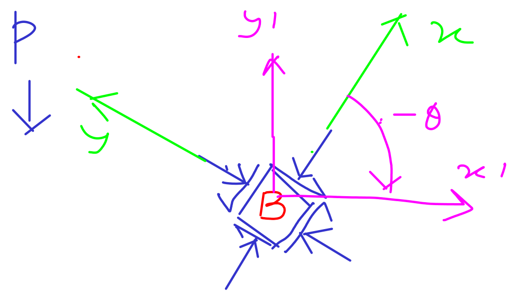
$$\sigma_{rr} = \frac{p}{2}\left[(1+K)\left(1-\frac{a^2}{r^2}\right) - (1-K)\left(1-4\frac{a^2}{r^2}+3\frac{a^4}{r^4}\right)\cos 2\theta\right]$$

$$\sigma_{\theta\theta} = \frac{p}{2}\left[(1+K)\left(1+\frac{a^2}{r^2}\right) + (1-K)\left(1+3\frac{a^4}{r^4}\right)\cos 2\theta\right]$$

$$\sigma_{r\theta} = \frac{p}{2}\left[(1-K)\left(1+2\frac{a^2}{r^2}-3\frac{a^4}{r^4}\right)\sin 2\theta\right]$$



What happens at ∞ at θ angle ?



$\forall\ r = \infty\ ,\ \theta = \theta$

$$\sigma_{rr} = \frac{p}{2}\left[(1+k) - (1-k)\cos 2\theta\right]$$

$$\sigma_{\theta\theta} = \frac{p}{2}\left[(1+k) + (1-k)\cos 2\theta\right]$$

$$\sigma_{r\theta} = \frac{p}{2}\left[(1-k)(1)(\sin 2\theta)\right]$$

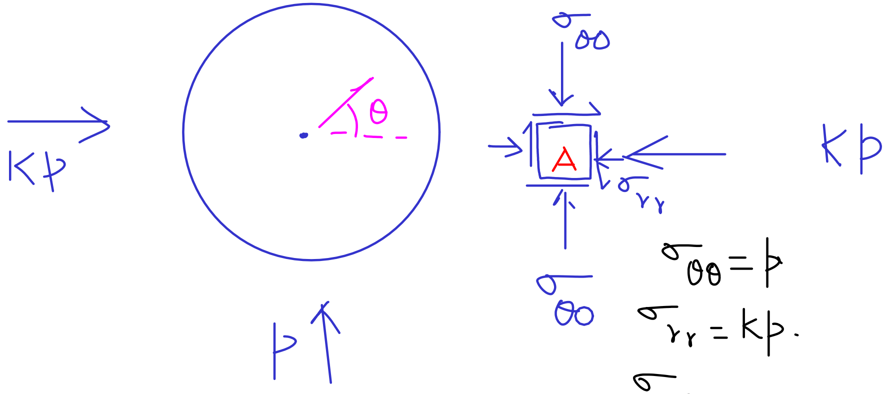
Let's do with example

Let $p = 2$ MPa, $k = 1.5$ & $\theta = 30^\circ$

$\sigma_{rr} = 2.75$ MPa

$\sigma_{\theta\theta} = 2.25$ MPa

$\sigma_{r\theta} = -\dfrac{\sqrt{3}}{4}$ MPa.



$\sigma_{\theta\theta} = p$

$\sigma_{rr} = kp.$

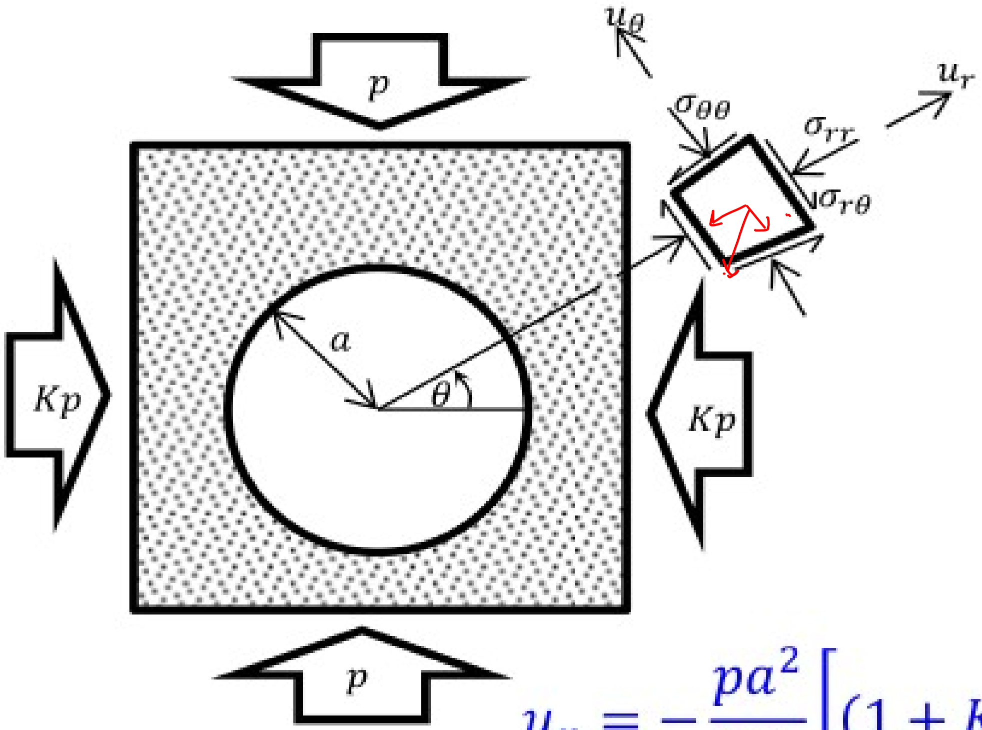
$\sigma_{r\theta} = 0$

$$\tau = \begin{bmatrix} 2.75 & -\frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} & 2.25 \end{bmatrix} \qquad L = \begin{bmatrix} \cos(-30) & \sin(-30) \\ -\sin(-30) & \cos(-30) \end{bmatrix}$$

or, $$L = \begin{bmatrix} \frac{\sqrt{3}}{2} & 0.5 \\ -0.5 & \frac{\sqrt{3}}{2} \end{bmatrix} \Rightarrow L^T = \begin{bmatrix} \frac{\sqrt{3}}{2} & -0.5 \\ 0.5 & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$\tau' = L\,\tau\,L^T = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

Matches the principal stress of the in-situ stress.

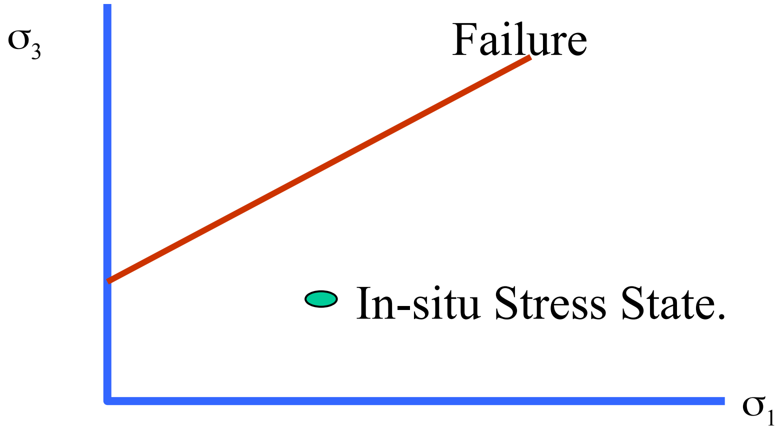


$$u_r = -\frac{pa^2}{4Gr}\left[(1+K) - (1-K)\left\{4(1-\nu) - \frac{a^2}{r^2}\right\}\cos 2\theta\right]$$

$$u_\theta = -\frac{pa^2}{4Gr}\left[(1-K)\left\{2(1-2\nu) + \frac{a^2}{r^2}\right\}\sin 2\theta\right]$$

Why measure in-situ Stresses ?

- Engineering analyses require boundary conditions
- One of the most **important** boundary conditions for the analysis of underground excavations is the **in-situ stress**.

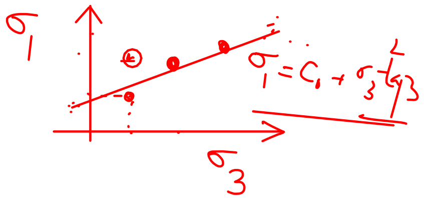


$$FOS = \frac{Strength}{Stress}$$

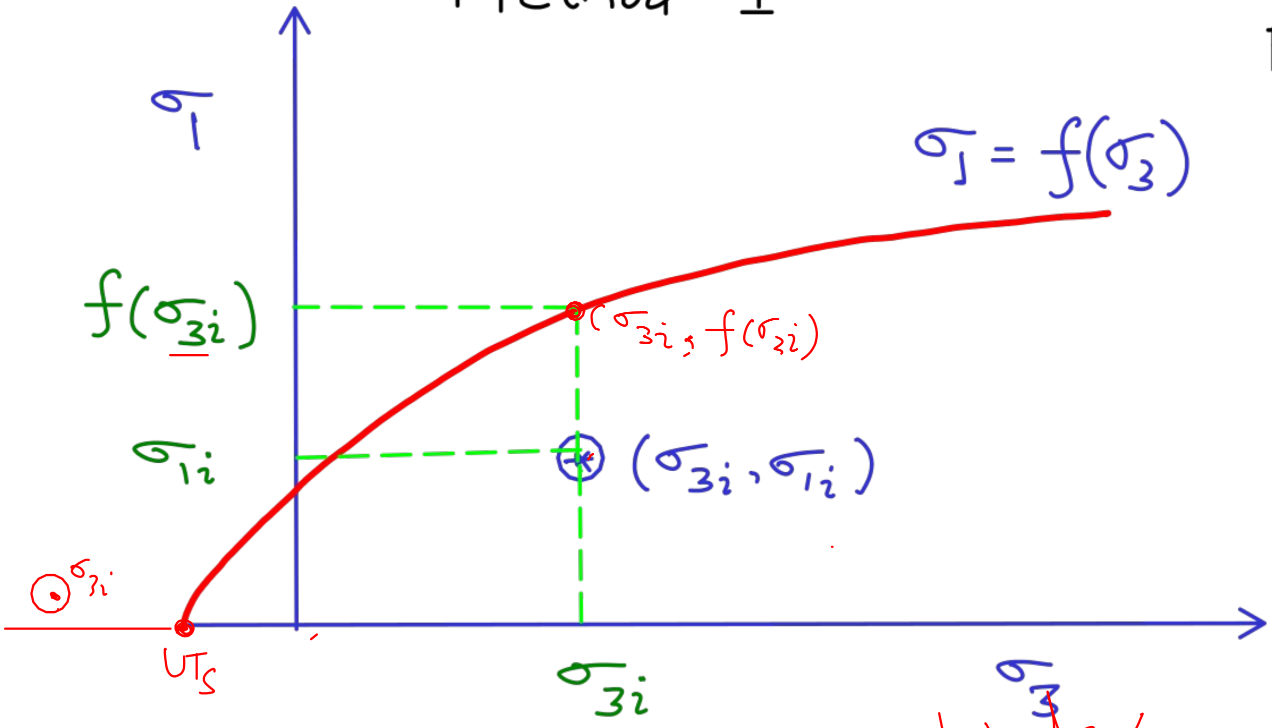
10t = Strength

5t = Stress

$F_{os} = \frac{10}{5} = 2$



Method – I



Not so popular

$$FOS = \frac{\text{Strength}}{\text{Stress}}$$

$$= \frac{f(\sigma_{3i})}{\sigma_{1i}}$$

However, when $-\sigma_{3i} > \sigma_t$

The

$$FOS = -\frac{\sigma_t}{\sigma_{3i}}$$